

FocusFlow

FREE · Windows

Product & Competitive Audit · v1.0.5 · May 2026

Source: full audit of 80 .kt source files + live competitor research

POSITIONING STATEMENT

For Windows users who need to actually stop using distracting apps — not just be reminded — FocusFlow is a free desktop blocker that kills processes in real time via Win32 APIs. Unlike Freedom (\$39.99/yr) or Cold Turkey Pro (\$39 one-time) it requires no subscription, blocks at the OS level including VPN bypass prevention, and includes a full kiosk mode that replaces your desktop.

AT A GLANCE

150

Windows users

12

Android users

19

Screens

10

DB tables

80+

Source files

100+

Safe whitelist

TOP 3 STRATEGIC RECOMMENDATIONS

1

Get on Google Play Store.

Android has 12 users because sideloading kills discoverability. A \$25 one-time fee is the only real path to Android growth. The Play Store is non-negotiable.

2

Add remote crash telemetry.

CrashReporter.kt writes crash logs to local disk only — you never see production crashes. Add a fire-and-forget HTTP POST (even a Discord webhook) on crash. You are currently blind.

3

Surface the VPN blocker — users do not know it exists.

VpnNetworkScreen is fully built and wired but never mentioned in changelog or onboarding. Blocking 35+ VPNs is a killer differentiator no competitor has. Market it immediately.

COMPETITIVE LANDSCAPE

	FocusFlow TBTechs	Cold Turkey getcoldturkey.com	Freedom freedom.to	RescueTime rescuetime.com
Price	FREE	\$39 one-time	\$39.99/yr	\$78/yr
Model	Free forever	Freemium	Subscription	Subscription
Platform	Windows + Android(sideload)	Windows + Mac	Mac/Win/iOS/Android/Linux	Mac/Win/Android/Chrome
Stage	150 users · solo dev	Bootstrapped · established	3.5M users · funded	Established · SMB-focused
	STRENGTHS ● Real Win32 process killing ● Focus Launcher kiosk mode ● VPN blocker (35+ clients) ● Free, no account needed WEAKNESSES ● Windows-only ● No Google Play Store ● No cloud sync ● No browser extension	STRENGTHS ● Hard-lock tamper-proof mode ● Allowlist mode ● Windows + Mac ● Strong community/Reddit WEAKNESSES ● Pro behind paywall ● Dated UI ● No mobile ● No kiosk/launcher	STRENGTHS ● Cross-device sync (6 platforms) ● Ambient focus sounds ● Polished UX ● Scheduled sessions WEAKNESSES ● Soft blocking only (no process kill) ● No kiosk mode ● Expensive ● Account required	STRENGTHS ● Automatic time tracking ● Team dashboards ● Detailed analytics ● Focus sessions WEAKNESSES ● Blocking is secondary ● Very expensive ● No kiosk mode ● Account + team-focused

FEATURE MATRIX (' FULL ' PARTIAL ' NONE)

Feature	Wt	FocusFlow	Cold Turkey	Freedom	RescueTime
Real process killing (Win32)	5	'	'	'	'
Free with full blocking	5	'	%Ñ	%Ñ	%Ñ
No account required	5	'	'	'	'
Focus Launcher / Kiosk mode	5	'	'	'	'
VPN bypass blocking	5	'	'	'	'
Nuclear mode (escape-route kill)	4	'	%Ñ	'	'
Registry lockdown (DisableTaskMgr)	4	'	%Ñ	'	'
Watchdog auto-restart	4	'	'	'	'
Emergency kill switch (5min/day)	4	'	'	'	'
Keyword blocker (window title)	3	'	'	'	'
Hosts file website blocking	3	'	'	%Ñ	%Ñ
Block schedules (recurring)	3	'	'	'	'
Daily allowances (time caps)	3	'	'	%Ñ	'
Pomodoro / custom timer	3	'	%Ñ	'	'
Session PIN gate	3	'	%Ñ	'	'
Task management (CRUD + recurring)	3	'	'	'	'
Habit tracker + streak grid	2	'	'	'	'
Weekly reports + analytics	2	'	'	%Ñ	'
Production-grade crash recovery	3	'	%Ñ	%Ñ	%Ñ
Cross-device sync	5	'	'	'	'
macOS / Linux support	4	'	'	'	'
Mobile app on Play Store	4	'	'	'	'
Browser extension	3	'	%Ñ	'	'
Team / family management	3	'	'	'	'
Cloud backup	2	'	'	'	'
WEIGHTED SCORE		77%	43%	39%	42%

CODE AUDIT — CONFIRMED STRENGTHS (READ FROM ACTUAL .KT FILES)

ProcessMonitor.kt	Dual-mode detection: WinEventHook (instant EVENT_SYSTEM_FOREGROUND) + 2000ms poll fallback. ConcurrentHashMap.compute() prevents double-kills when hook+poll fire within cooldown window simultaneously.
NuclearMode.kt	3-layer escape enforcement: (1) single tasklist /FO CSV scan — down from ~8000 process spawns/min to 1 per tick, (2) batch taskkill with all targets, (3) NetworkBlocker outbound firewall rules per escape process.
KillSwitchService.kt	5-min daily emergency break budget. Persisted to DB with reset-date check. Auto-deactivates on exhaust. FocusLauncherService.onKillSwitchActivated/Deactivated() correctly re-engages kiosk.
VpnBlocker.kt	35+ named VPN processes blocked: NordVPN, ExpressVPN, ProtonVPN, WireGuard, Tor, Cisco AnyConnect, Palo Alto GlobalProtect, Fortinet FortiClient, Pulse Secure. Custom additions stored in DB. No competitor has this.
RegistryLockdown.kt	DisableTaskMgr + NoLogOff (HKCU, no admin) + HideFastUserSwitching (HKLM, silently skipped if not elevated). Calls UpdatePerUserSystemParameters() to refresh policies instantly. Cleanup on disable deletes keys entirely.
CrashReporter.kt	Production-grade: all threads + AWT EDT + coroutines covered. Pre-allocated 1MB OOM reserve freed first. AtomicBoolean re-entrancy guard. 4 write destinations (Desktop !' ~/.focusflow !' %TEMP% !' stderr). Full thread dump + GC info + DB WAL check in
ProcessMonitor.kt	100+ safe-process whitelist across 7 categories: core kernel (csrss/lsass/dwm), input stack (ctfmon/tabtip/textinputhost/osk/wisptis), 7 OEM peripheral suites (Synaptics/Elan/Logitech/Razer/SteelSeries/Corsair/ASUS ROG).
FocusLauncherService.kt	Shutdown hook in init{} block — taskbar restored even on SIGTERM/OOM. StateFlow for isActive/breakActive/isHardLocked/canTakeBreak. Break budget is per-calendar-day checked via DB date string.
FocusSessionService.kt	UUID session IDs, @Synchronized start() prevents race on rapid UI taps, milestone events fired at 25/50/75% elapsed, notes field flushed to DB on end(), lastSummary StateFlow for post-session UI.
DailyAllowanceTracker.kt	Polls foreground every 10s, accumulates usage, kills + injects into ProcessMonitor.dailyAllowanceBlockedProcesses for instant WinEventHook coverage once cap exceeded. Midnight reset via date comparison.
Database.kt	TARGET_VERSION=5. isReady guard on all get/set calls. 10 tables: tasks, focus_sessions, block_rules, block_schedules, daily_allowances, daily_notes, temptation_log, habits, network_cutoff_rules, custom_block_presets.
General	7+ locale i18n (Translations.kt). AutoBackupService rolling local backups. WatchdogInstaller via Task Scheduler every 2 minutes. 19 screens all wired in App.kt navigation. Zero TODO/FIXME comments in codebase.

CODE AUDIT — CONFIRMED WEAKNESSES / BUGS / RISKS (FROM .KT FILES)

CRITICAL	NuclearMode.kt / RegistryLockdown.kt / KillSwitchService.kt — catch(_: Exception){} swallows ALL failures silently. If taskkill fails, registry write fails, or DB write fails nothing is logged. Enforcement may be broken with no indication.
CRITICAL	CrashReporter.kt — Crash logs written ONLY to local disk (Desktop / ~/.focusflow / %TEMP%). Developer never sees production crashes. Zero telemetry = zero visibility into real user failures.
HIGH	ProcessMonitor.kt — launcherSafeProcesses comment reads "Last reviewed: 2025-05" (now 2026). Hardcoded list of 100+ OEM drivers. New peripherals not covered until manual code update.
HIGH	DailyAllowanceTracker.kt — Polls every 10,000ms. Up to 10 seconds of unblocked usage after cap exceeded before kill fires. Not real-time. WinEventHook path is instant by contrast.
HIGH	VpnNetworkScreen.kt — Fully built, wired in navigation (Screen.VPN_NETWORK), has DB table. Never mentioned in changelog, onboarding, or store description. Users have no idea it exists.
HIGH	RegistryLockdown.kt — Code comment: "Ctrl+Alt+Del cannot be blocked from user mode — kernel-level Secure Attention Sequence." Kiosk can still be interrupted. This is a documented hard limitation.
HIGH	RegistryLockdown.kt — HideFastUserSwitching requires HKLM write (admin). Silently skipped without admin. Fast user switching remains available on most standard (non-admin) installs.
MEDIUM	ProcessMonitor.kt — LAUNCHER_SWEEP_INTERVAL_MS = 250ms in kiosk mode. 4 full process scans per second. May cause sustained CPU spikes on low-end or older laptops during long kiosk sessions.
MEDIUM	VpnBlocker.kt — isVpnProcess() calls Database.getSetting("vpn_block_enabled") on EVERY WinEventHook event (hot-path). No caching. Unnecessary synchronous DB read per foreground change.
MEDIUM	No browser extension. Hosts-file blocking bypassed by changing DNS server or enabling DoH (DNS-over-HTTPS) in browser settings. Extension-level blocking would be more robust for website blocking.
MEDIUM	No cloud backup. DB lives in ~/.focusflow/focusflow.db only. Hard drive failure or OS reinstall = total data loss. BackupService does local rolling backups only — useless if disk fails.
LOW	Non-Windows: OsBanner shows a warning but JNA services still attempt initialization. User32/Advapi32 calls throw on Linux/macOS. Silent failures, not clean hard errors with meaningful messages.
LOW	Android app not on Google Play Store. Sideload + AppGallery (Huawei-only) = invisible to 99%+ of Android users. Direct consequence: 12 Android users vs 150 Windows users.

WHITE SPACE — WHAT NO COMPETITOR DOES WELL

Free + Real Enforcement

Freedom and RescueTime charge \$40-80/yr for soft blocking. Cold Turkey Pro charges \$39 one-time. FocusFlow is the ONLY tool combining zero cost with actual Win32 process killing. Lead every message with this.

VPN Bypass Prevention

No other consumer focus app kills VPN clients during a session. FocusFlow blocks 35+ VPN processes including corporate VPNs (Cisco AnyConnect, Palo Alto GlobalProtect). This is a Kano Delighter — zero competitors have it.

Desktop Kiosk Mode

Cold Turkey has a "frozen turkey" mode but no kiosk replacing the desktop with a custom workspace. Freedom has nothing equivalent. The most extreme focus tool feature available in any free app.

Integrated Task Management

Cold Turkey and Freedom do zero task management. FocusFlow has full CRUD tasks, recurring tasks, alarms, and per-task blocked-app lists. Differentiated from every competitor.

ACTION PLAN

1 IMMEDIATELY**Add remote crash telemetry**

CrashReporter.kt has SUPPORT_EMAIL = support@tbtechs.dev but zero network calls. Add a fire-and-forget HTTP POST of crash summary (no PII) on crash. A Discord webhook is enough. You are currently completely blind to all user crashes.

2 THIS WEEK**Surface VPN blocking to users**

Add VPN blocker entry to v1.0.5 changelog. Add mention in onboarding dialog. Update Microsoft Store description. This feature exists but zero users know about it — it is the most unique thing in the app.

3 THIS WEEK**Fix VpnBlocker hot-path DB read**

VpnBlocker.isVpnProcess() calls Database.getSetting() on every WinEventHook event. Cache enabled state in @Volatile Boolean, refresh on settings save. Eliminates unnecessary DB reads in the enforcement hot-path.

4 THIS MONTH**Submit Android to Google Play Store**

\$25 one-time. Android without Play Store is effectively invisible. You have 12 Android users vs 150 Windows — Play Store is the only lever. Frame app carefully for review (user-controlled, user-consent framing).

5 NEXT RELEASE**Fix silent catch blocks**

Replace catch(_:Exception){} in NuclearMode, RegistryLockdown, KillSwitchService with at minimum a log to DB error table or CrashReporter call. Silent failures make critical enforcement layers completely undebuggable.

SOURCES

- [1] ProcessMonitor.kt (689 lines) — dual-mode enforcement, safe-process whitelist, all block-source injection points
- [2] NuclearMode.kt (257 lines) — 3-layer escape enforcement, escapeProcesses set, batch kill, firewall lock
- [3] KillSwitchService.kt (101 lines) — 5-min daily budget, DB persistence, FocusLauncherService integration
- [4] VpnBlocker.kt (113 lines) — 35+ VPN process list, custom process DB storage, isVpnProcess() hot-path issue confirmed
- [5] RegistryLockdown.kt (112 lines) — DisableTaskMgr/NoLogOff/HideFastUserSwitching, Ctrl+Alt+Del limitation documented in code
- [6] CrashReporter.kt (500+ lines) — thread coverage, OOM reserve, 4 write destinations, local-only telemetry gap confirmed
- [7] FocusLauncherService.kt (428 lines) — shutdown hook, break budget, hard-lock mode, kiosk StateFlows
- [8] FocusSessionService.kt (240 lines) — UUID sessions, milestone tracking, @Synchronized start()
- [9] DailyAllowanceTracker.kt (152 lines) — 10s polling confirmed, midnight reset, ProcessMonitor injection
- [10] Database.kt — TARGET_VERSION=5, isReady guard, 10 tables, versioned migrations confirmed
- [11] build.gradle.kts — version 1.0.5, Kotlin 1.9.22, Compose Multiplatform 1.6.1, JNA 5.14.0
- [12] Cold Turkey — getcoldturkey.com (live, May 2026) — free + Pro \$39 one-time; "most locked-down...difficult to circumvent"
- [13] Freedom — freedom.to (live, May 2026) — \$39.99/yr or \$199 lifetime; cross-device sync; 3.5M+ users
- [14] RescueTime — rescuetime.com (live, May 2026) — automatic time tracking; subscription-based; \$78/yr Focus plan